Proposed Solution

Date	17 September 2026
Team ID	SWTID-2026-3976
Project Title	AI FAQ Assistant API - AI-powered FAQ and customer support automation
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The AI FAQ Assistant is a secure REST API that lets support teams create, organise and search FAQs and use Google Gemini to draft answers and complete FAQ entries. It cuts repetitive writing effort, keeps content in one protected place and lets customers find answers quickly.

Project Overview

Objective	Automate FAQ creation and retrieval with a secure, AI-assisted backend so support teams spend less time on repetitive documentation.
Scope	User registration and login (JWT), FAQ CRUD, category handling, keyword search, AI answer and AI FAQ generation, centralised error handling, API testing with Postman.

Problem Statement

Description	Support teams handle large backlogs of repeated questions and struggle to organise entries, classify topics and write precise answers. Private documentation also lacks permission layers.
Impact	Solving this reduces manual effort and response delays, keeps documentation accurate and secure, and improves customer self-service.

Proposed Solution

Approach	A layered Node.js/Express API with Mongoose models, JWT authentication and a Gemini service module. AI output is generated on request and saved as an FAQ only when the author chooses.
Key Features	- JWT authentication with bcrypt password hashing - FAQ create, read, update and delete with author ownership - Keyword search across FAQs - AI answers and AI-generated FAQs using Google Gemini - Input validation and centralised error handling

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	Processor	Intel Core i5 (8th Gen) / AMD Ryzen 5 or better; no GPU needed
Memory	RAM	8 GB minimum (16 GB recommended)
Storage	Disk space	1 GB free
Software		
Frameworks	Backend frameworks	Node.js 18+, Express.js 4/5
Libraries	Additional packages	mongoose, dotenv, cors, jsonwebtoken, bcrypt, @google/genai
Development Environment	IDE and API tools	Visual Studio Code, Postman / ThunderClient, npm 9+
Data		

Resource Type	Description	Specification / Allocation
Data	Source, size, format	Data created by users through the API and stored in MongoDB (JSON documents); no external dataset required